CS 3244

Machine Learning

Week 8 Pre-Class

Neural Networks

8 pre

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Ang Yi Zhe

Background Photo credits: Rafiq Mirza, Luan Ahn, Rosmarie Voegtli @ Flickr

Recap and Forecast

Understand how validation and regularization complement to address overfitting:

- **Regularization:** a means of restraining the model.
  - L^p norm forms of regularization.

$$J_{test}(h) = J_{train}(h) + \text{overfit penalty}$$

- **Validation**, as a reality check by peeking (at the bottom line).
  - Cross Validation variants: k-fold or leave one out.

Learning outcomes for this week

- Describe the basic idea of Artificial Neural Networks (ANN or NN):
  - Explain how neural networks combine linear units with a non-linear activation function;
  - Explain the topology of a NN;
- Name potential problems with NNs;
- Describe how a neural network is trained using the backpropagation algorithm;
- Name NN regularization techniques;
- Obtain practical experience constructing NNs using PyTorch.

Revisiting XOR

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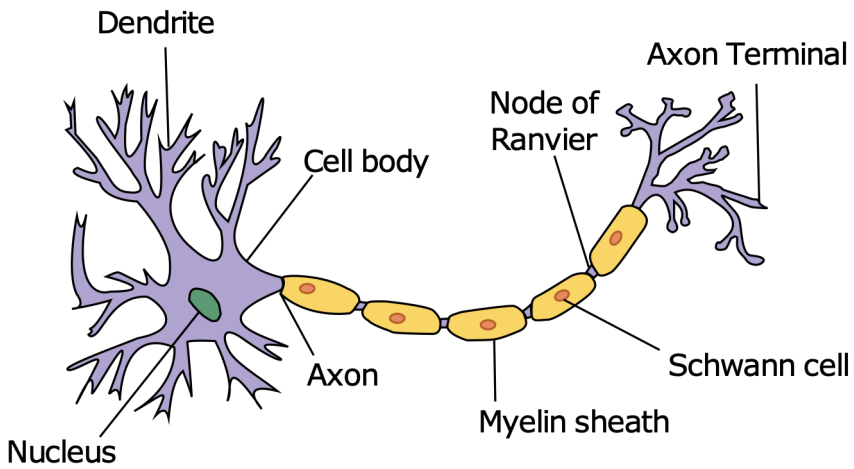
Biological Inspiration



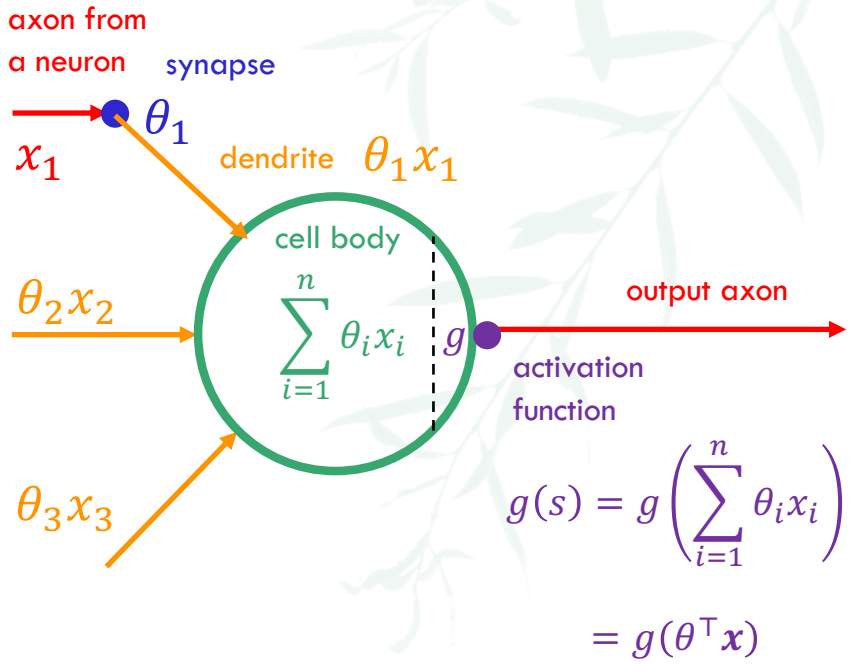
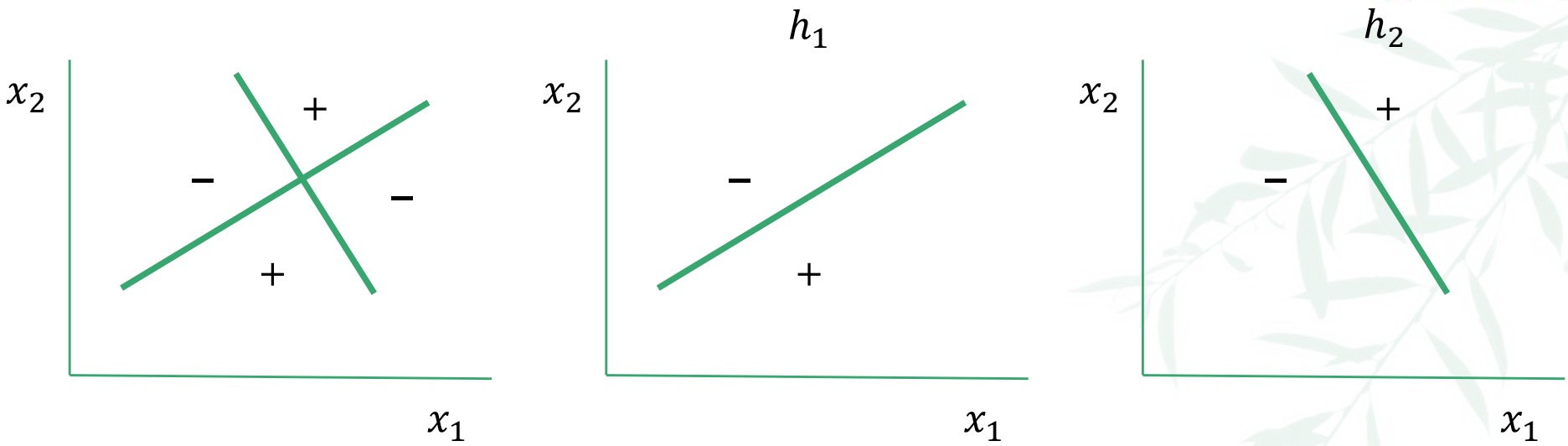
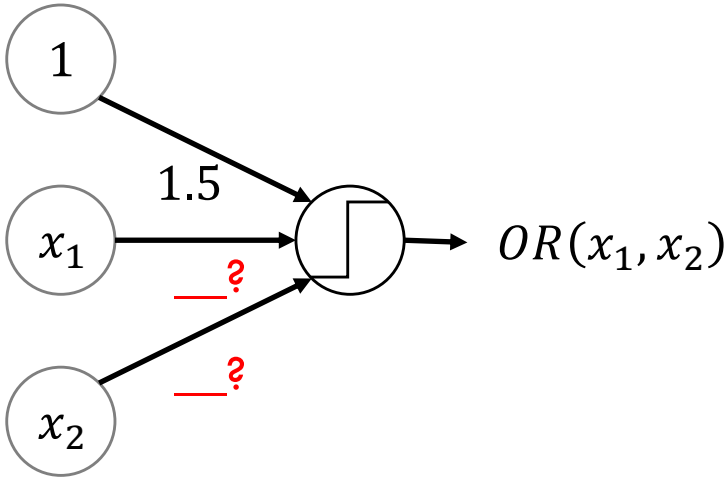


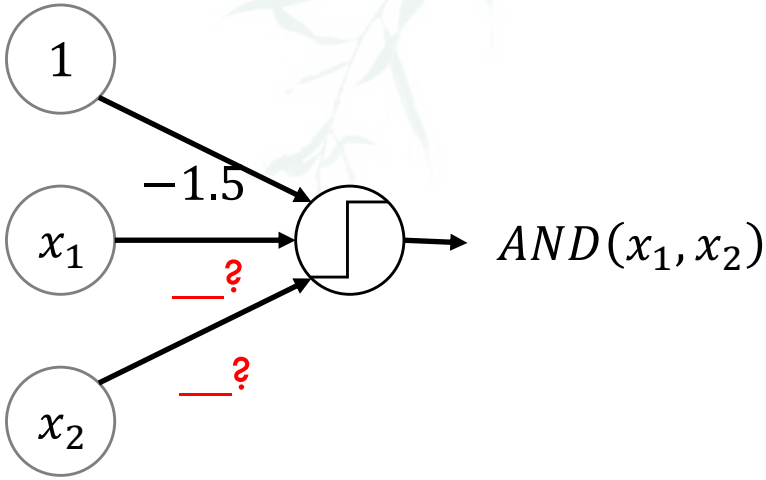
Diagram credits: Dhp1080 - Own work, CC BY-SA 3.0 via Wikimedia Commons.

Combining Perceptrons

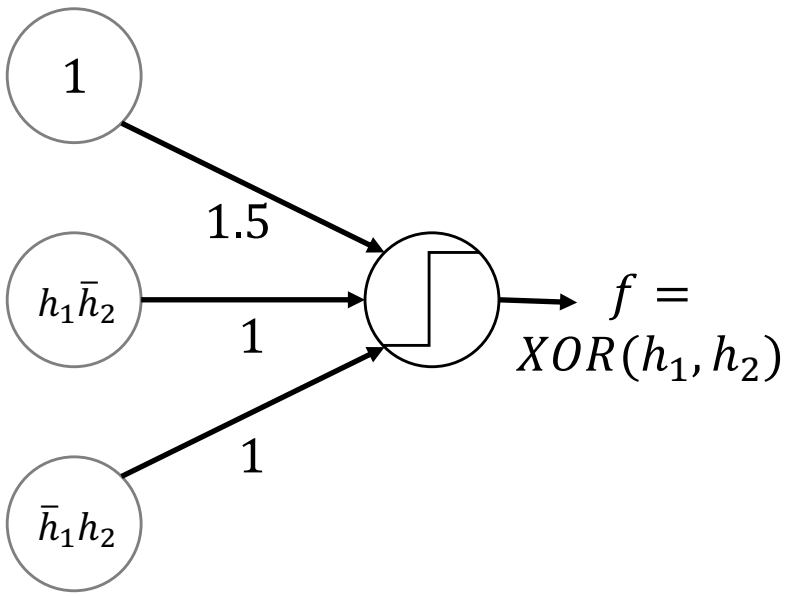


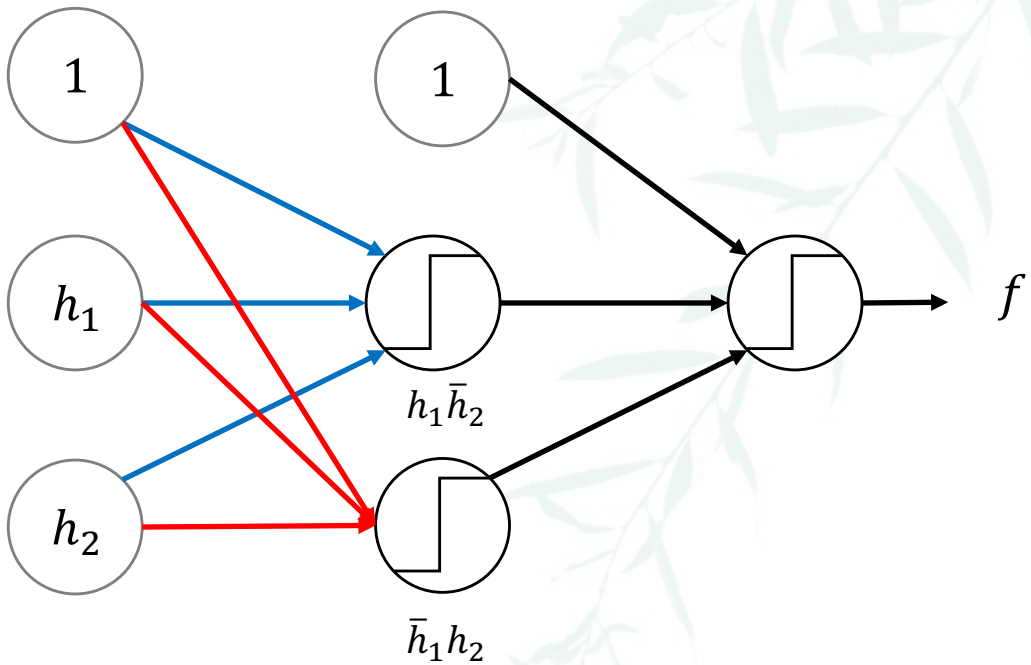
$$\text{sign}(\theta^\top \boldsymbol{x}) \qquad y = \pm 1$$





Creating Layers





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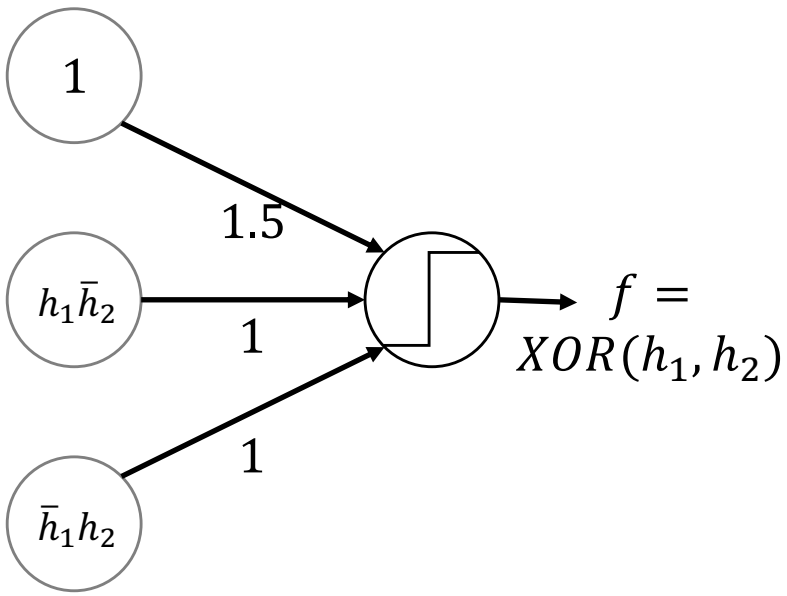
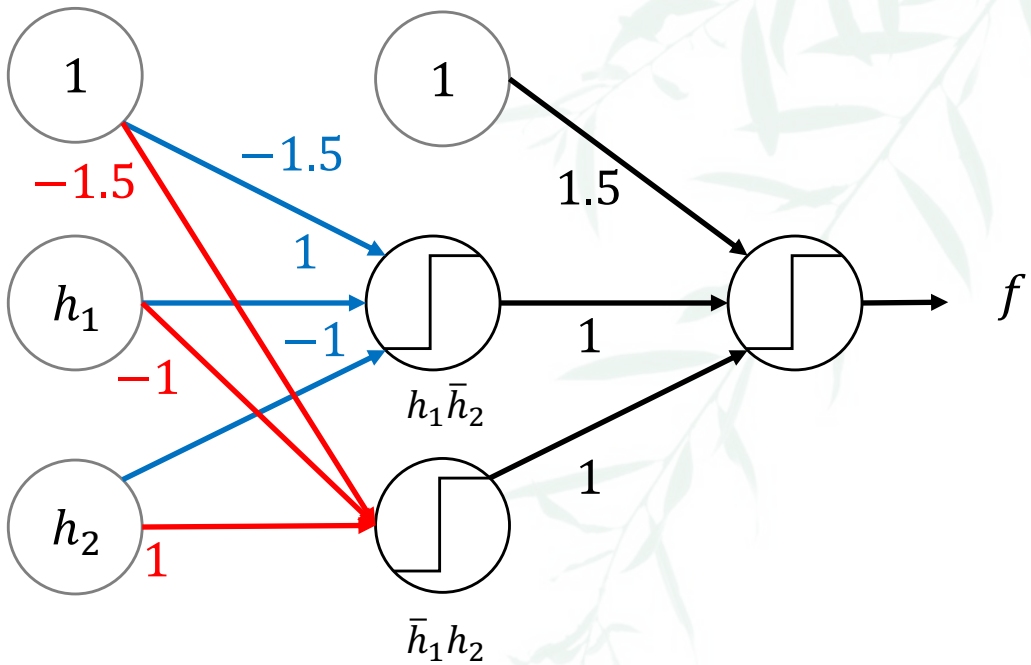
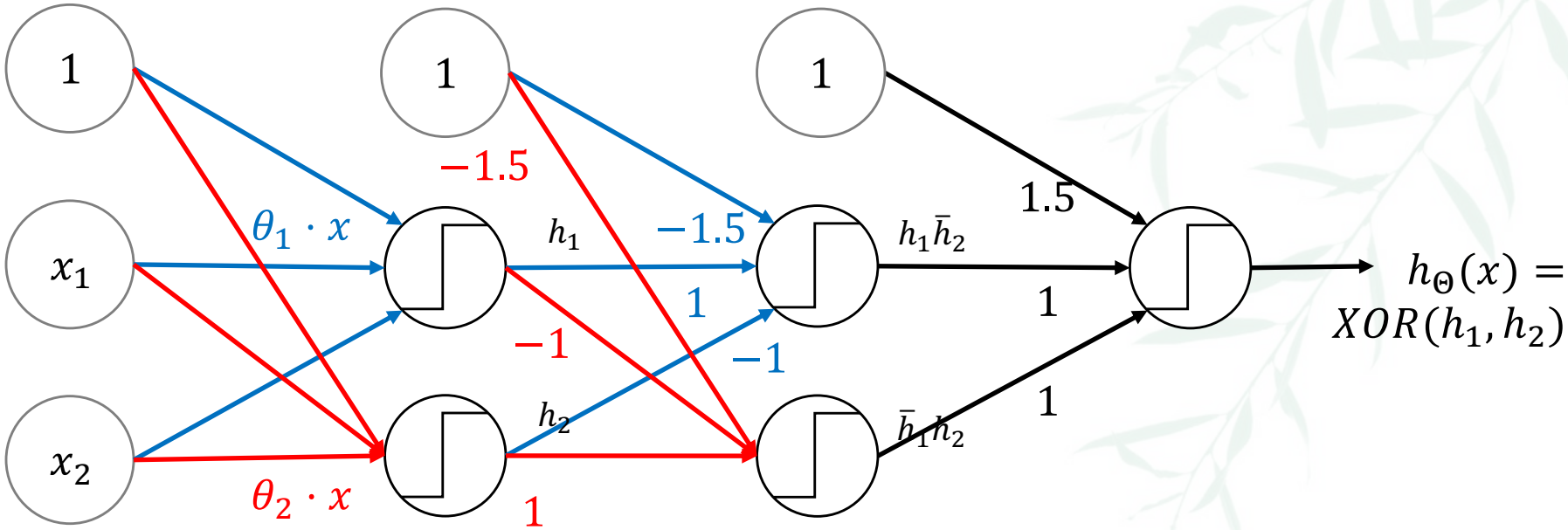




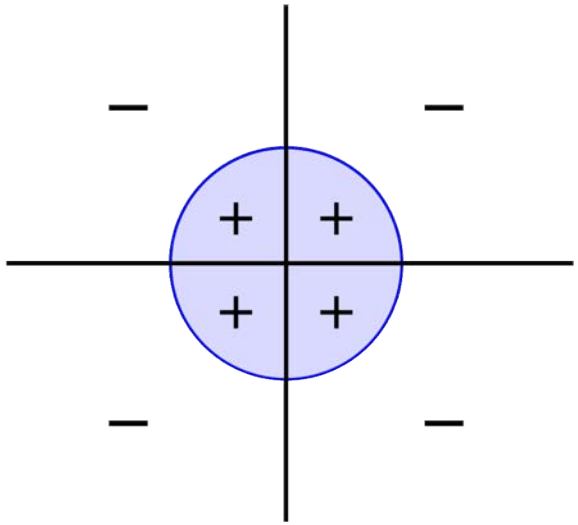
The Multilayer Perceptron

3 layers

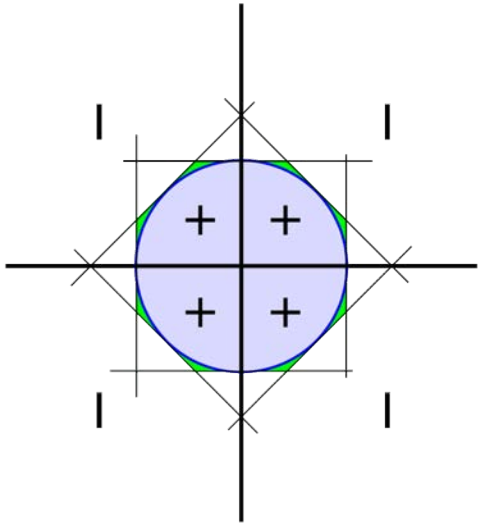
"feedforward" ≡ no loops



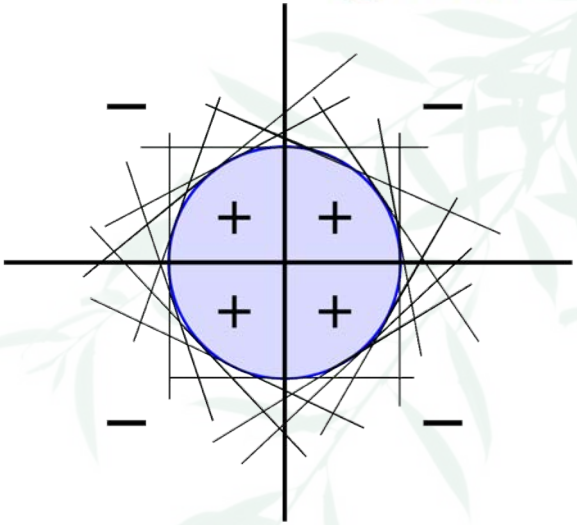
A Powerful Model



Target



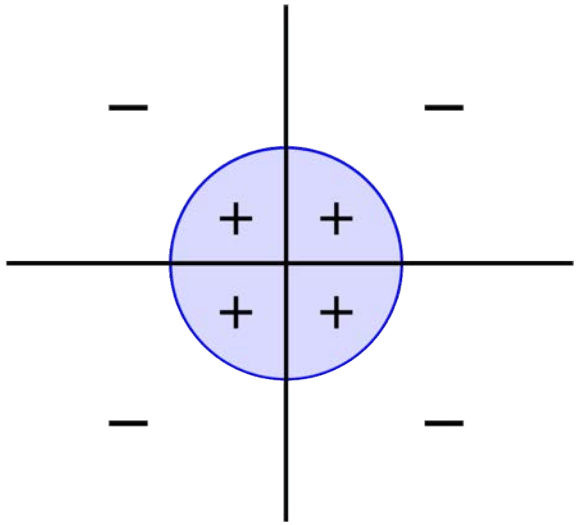
8 perceptrons



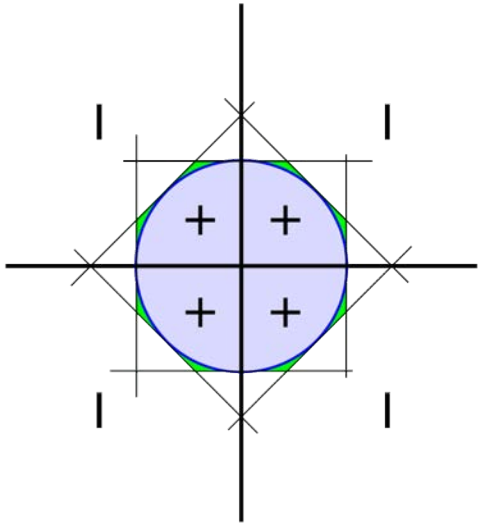
16 perceptrons

2 red flags:

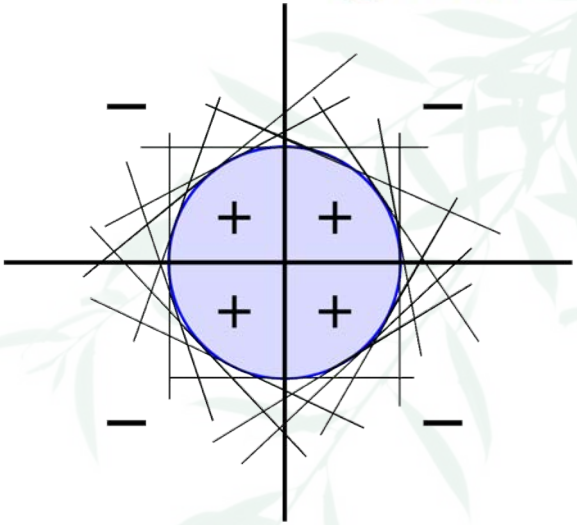
A Powerful Model



Target



8 perceptrons



16 perceptrons

2 red flags:

for generalization and

for optimization

NN Topology

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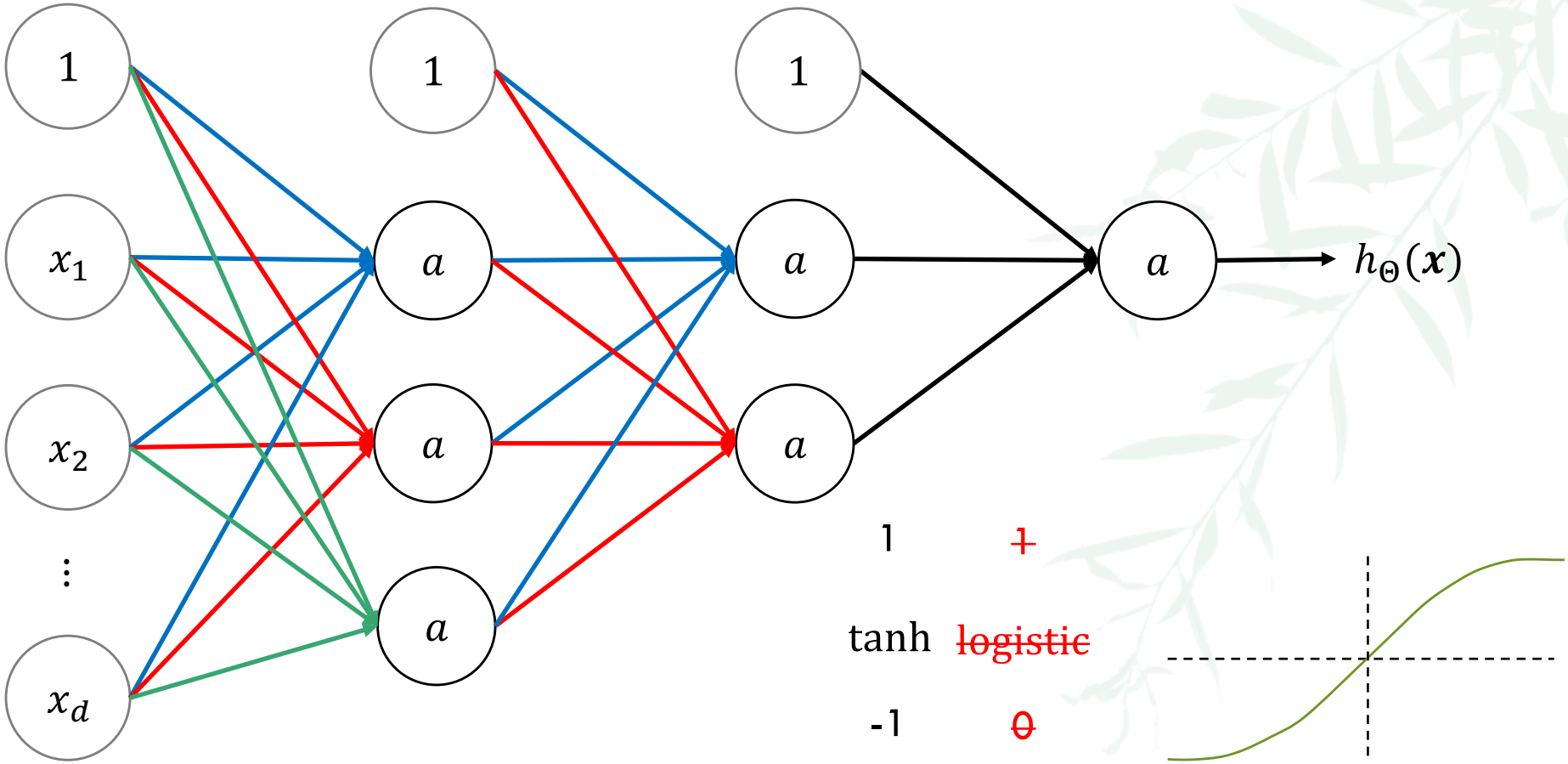
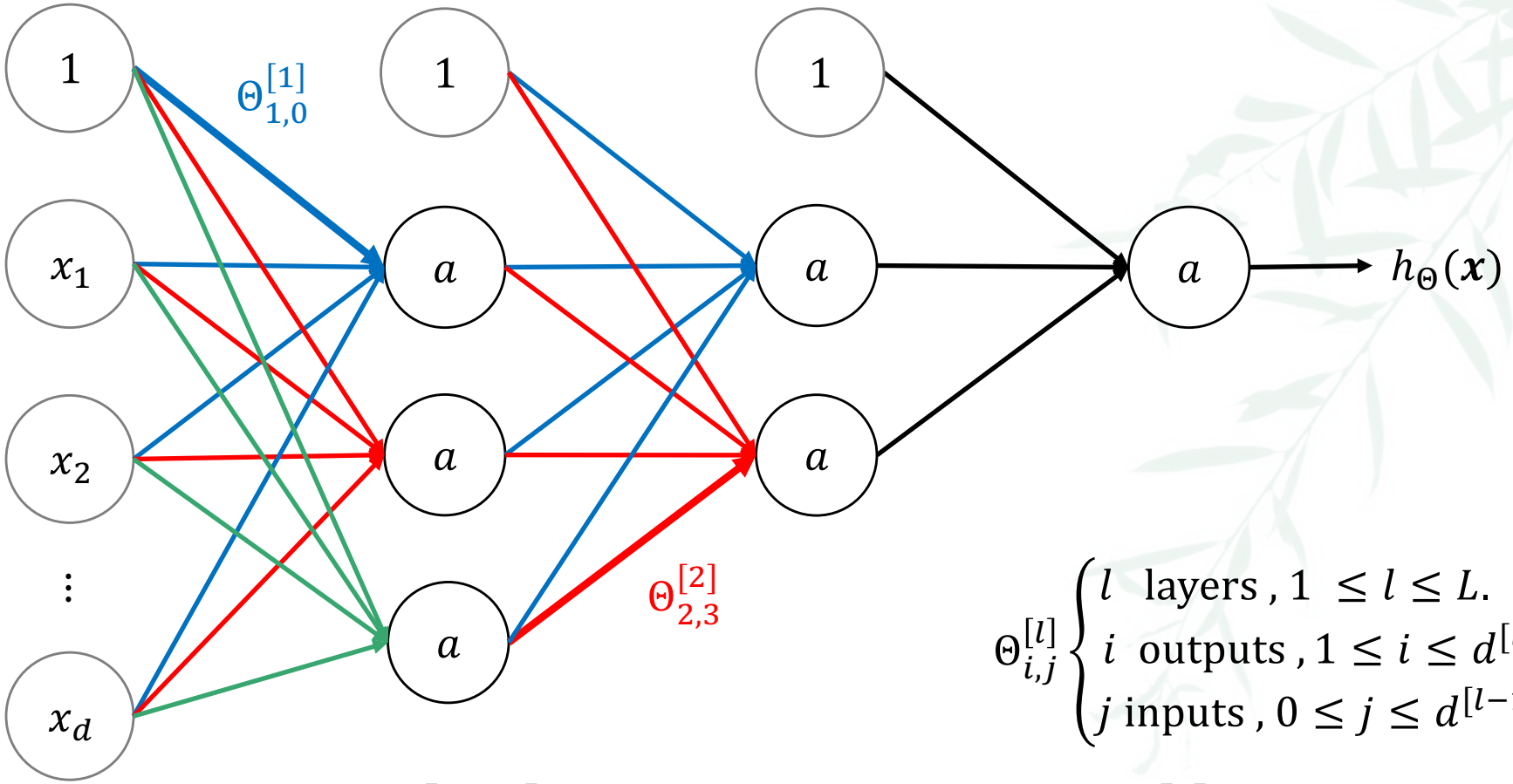
The Neural Network



Input $\boldsymbol{x}$ Hidden layers $1 \leq l < L$ Output layer $l = L$

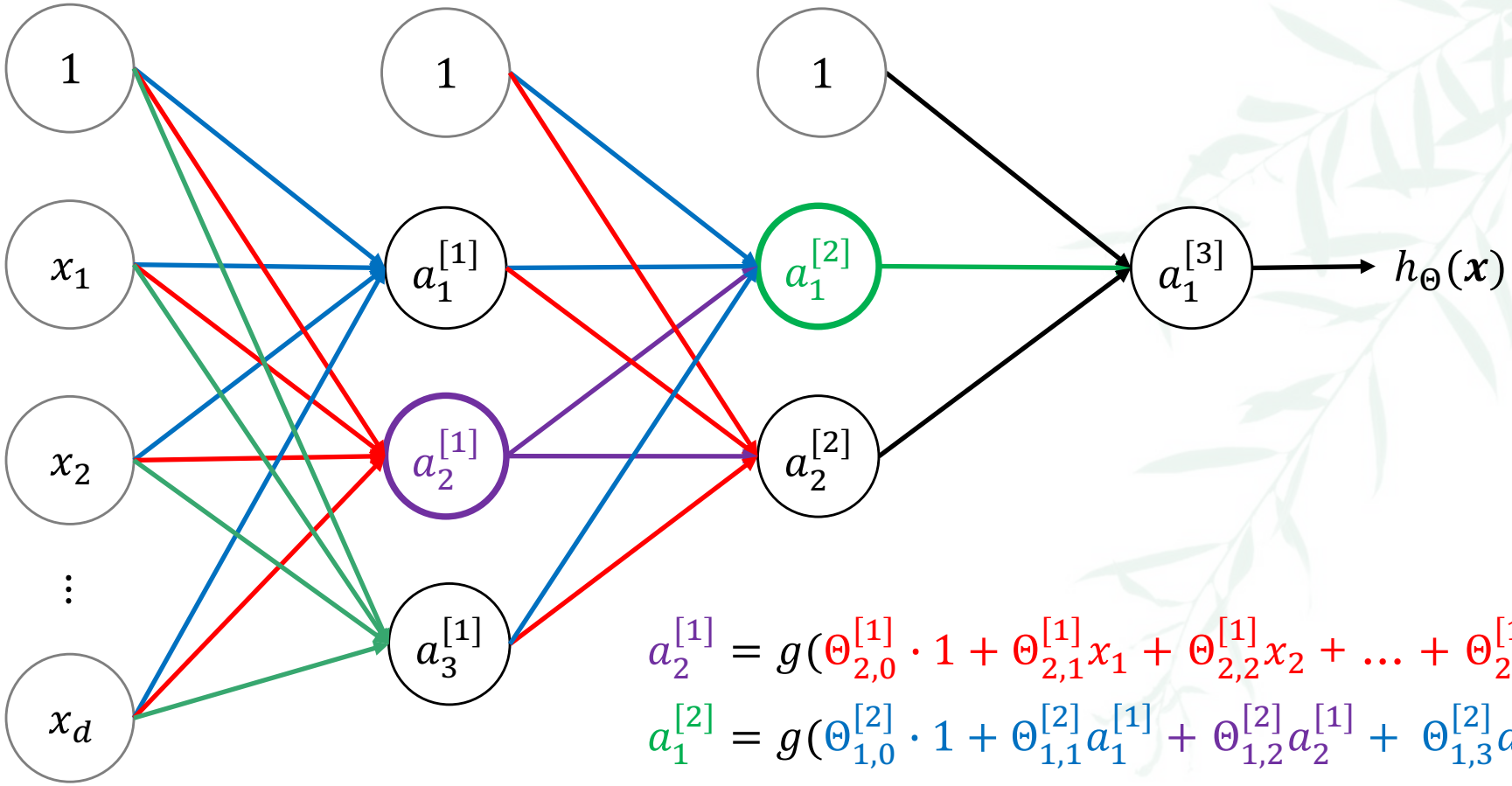
NN Indices



$$\Theta_{i,j}^{[l]} \begin{cases} l \text{ layers}, 1 \le l \le L. \\ i \text{ outputs}, 1 \le i \le d^{[l]} \\ j \text{ inputs}, 0 \le j \le d^{[l-1]} \end{cases}$$

If a network has $d^{[l-1]}$ units in layer $l-1$, $d^{[l]}$ units at layer l, then $\Theta^{[l]}$ will be of dimension $(d^{[l-1]}+1)\times d^{[l]}$.

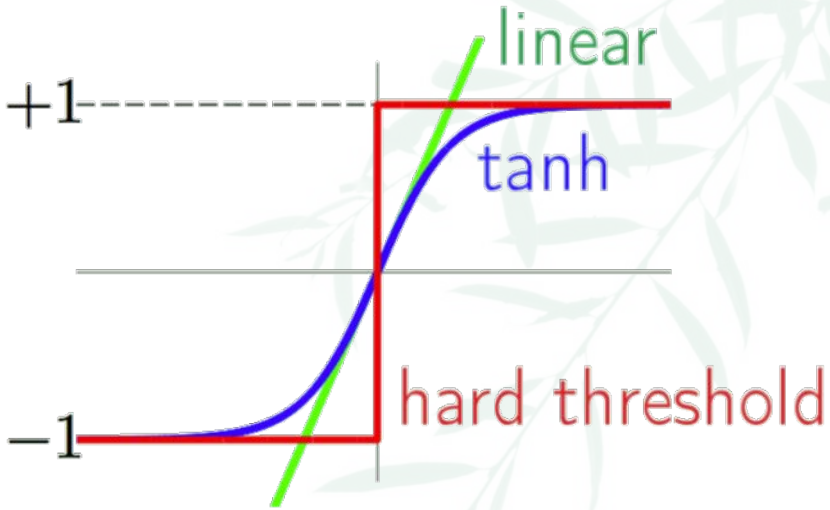
NN Activations



$$a_2^{[1]} = g(\Theta_{2,0}^{[1]} \cdot 1 + \Theta_{2,1}^{[1]} x_1 + \Theta_{2,2}^{[1]} x_2 + \ldots + \Theta_{2,d}^{[1]} x_d)$$

$$a_1^{[2]} = g(\Theta_{1,0}^{[2]} \cdot 1 + \Theta_{1,1}^{[2]} a_1^{[1]} + \Theta_{1,2}^{[2]} a_2^{[1]} + \Theta_{1,3}^{[2]} a_3^{[1]})$$

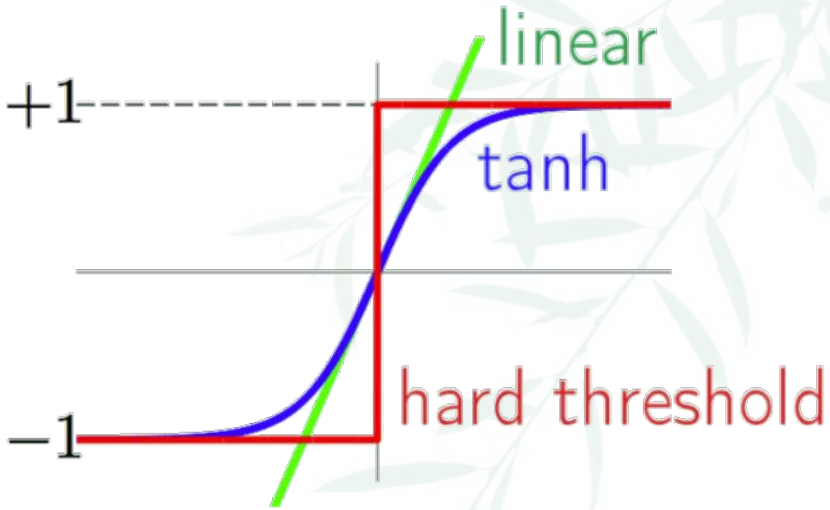
How the Network Operates



$$g(s) = \tanh(s) = \frac{e^s - e^{-s}}{e^s + e^{-s}}$$

How the Network Operates

$$\Theta_{i,j}^{[l]} \begin{cases} 1 \le l \le L & \text{layers} \\ 1 \le i \le d^{[l]} & \text{outputs} \\ 0 \le j \le d^{[l-1]} & \text{inputs} \end{cases}$$



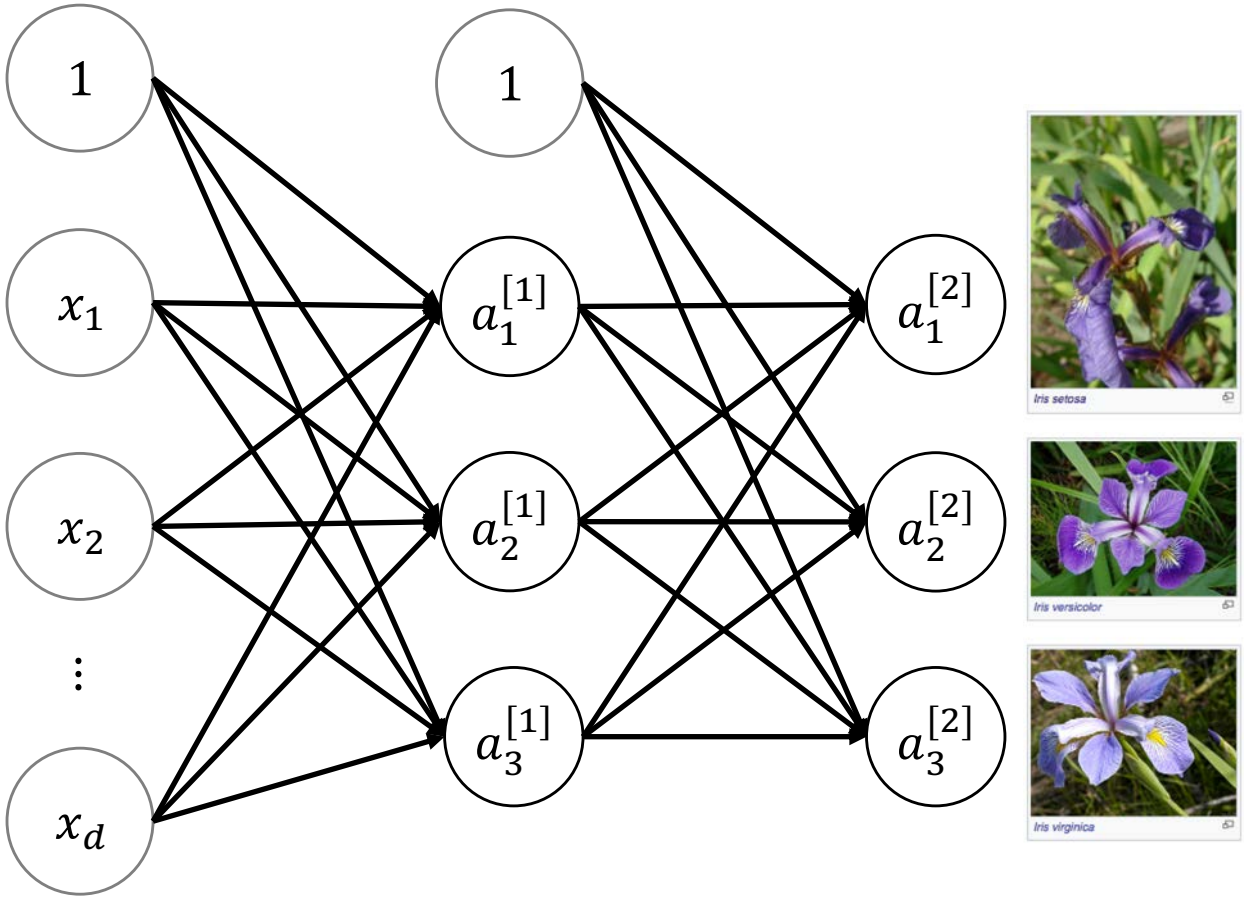
$$x_i^{[l]} = a_i^{[l]} = g\left(s_i^{[l]}\right) = g\left(\sum_{j=0}^{d^{[l-1]}} \Theta_{i,j}^{[l]} x_j^{[l-1]}\right)$$

$$= g((\Theta_i^{[l]})^\top \boldsymbol{x}^{[l-1]})$$

$$g(s) = \tanh(s) = \frac{e^s - e^{-s}}{e^s + e^{-s}}$$

Feedforward: Apply $\boldsymbol{x}$ to $x_1^{[0]} \dots x_{d^{[0]}}^{[0]} \to\to x_1^{[L]} = h(\boldsymbol{x})$

Multiclass classification



Multiple output units:
$h_\Theta(\boldsymbol{x}) \in \mathbb{R}^3$

Want $h_\Theta(\boldsymbol{x}) \approx \begin{bmatrix} 1 \\ -1 \\ -1 \end{bmatrix}$ when *setosa*, $h_\Theta(\boldsymbol{x}) \approx \begin{bmatrix} -1 \\ 1 \\ -1 \end{bmatrix}$ when *versicolor* and $h_\Theta(\boldsymbol{x}) \approx \begin{bmatrix} -1 \\ -1 \\ 1 \end{bmatrix}$ when *virginica*.

Diagram credits: Frank Mayfield, Danielle Langlois & Radomił Binek - Own work, CC BY-SA 3.0 via Wikimedia Commons.